

MAST32001 Computational Statistics - Lecture 9

Bayesian Statistics and MCMC Foundations

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From likelihood to posterior

Why Bayesian statistics here?

Bayesian inference is based on the posterior distribution. For data y and parameter θ ,

$$p(\theta | y) = \frac{p(y | \theta)p(\theta)}{p(y)} \propto p(y | \theta)p(\theta).$$

The ingredients are:

- ▶ $p(y | \theta)$: likelihood;
- ▶ $p(\theta)$: prior;
- ▶ $p(\theta | y)$: posterior;
- ▶ $p(y) = \int p(y | \theta)p(\theta) d\theta$: marginal likelihood.

The computational challenge is that the posterior may be difficult to normalize, integrate, or sample from directly.

Likelihood: start from the sampling model

Suppose

$$Y_1, \dots, Y_n \mid \mu, \sigma^2 \stackrel{iid}{\sim} N(\mu, \sigma^2),$$

with σ^2 known.

For one observation,

$$p(y_i \mid \mu) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left\{ -\frac{(y_i - \mu)^2}{2\sigma^2} \right\}.$$

Independence gives the joint density

$$p(y_1, \dots, y_n \mid \mu) = \prod_{i=1}^n p(y_i \mid \mu).$$

Viewed as a function of μ , this is the likelihood:

$$L(\mu; y) \propto \exp \left\{ -\frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - \mu)^2 \right\}.$$

Likelihood: simplify the quadratic

Use

$$\sum_{i=1}^n (y_i - \mu)^2 = \sum_{i=1}^n (y_i - \bar{y})^2 + n(\bar{y} - \mu)^2.$$

The first term does not depend on μ , hence

$$L(\mu; y) \propto \exp \left\{ -\frac{n}{2\sigma^2} (\mu - \bar{y})^2 \right\}.$$

Thus, as a function of μ , the likelihood has a normal kernel:

$$L(\mu; y) \propto \exp \left\{ -\frac{1}{2(\sigma^2/n)} (\mu - \bar{y})^2 \right\}.$$

This derivation is worth learning: **sampling model** $\rightarrow$ **product** $\rightarrow$ **likelihood** $\rightarrow$ **log/quadratic simplification**.

Conjugate Bayesian update

Take

$$\mu \sim N(\mu_0, \tau_0^2).$$

Then

$$p(\mu \mid y) \propto \exp \left\{ -\frac{n}{2\sigma^2} (\mu - \bar{y})^2 \right\} \exp \left\{ -\frac{1}{2\tau_0^2} (\mu - \mu_0)^2 \right\}.$$

Collecting the quadratic terms gives

$$\mu \mid y \sim N(\mu_n, \tau_n^2),$$

where

$$\tau_n^2 = \left(\frac{1}{\tau_0^2} + \frac{n}{\sigma^2} \right)^{-1}, \quad \mu_n = \tau_n^2 \left(\frac{\mu_0}{\tau_0^2} + \frac{n\bar{y}}{\sigma^2} \right).$$

Posterior mean as weighted information

Write the posterior mean as

$$\mu_n = w\mu_0 + (1 - w)\bar{y},$$

with

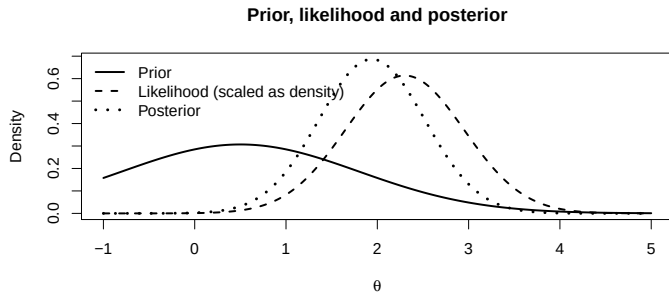
$$w = \frac{1/\tau_0^2}{1/\tau_0^2 + n/\sigma^2}.$$

The weights are determined by **precision**:

- ▶ prior precision: $1/\tau_0^2$;
- ▶ data precision: n/σ^2 .

As n grows, the data receive more weight.

Prior, likelihood and posterior: picture



The likelihood pulls toward values supported by the data; the prior contributes information before seeing the data; the posterior combines both.

Bayesian inference is more than a point estimate

If we can generate draws $\theta^{(1)}, \dots, \theta^{(M)}$ from the posterior, many quantities become Monte Carlo summaries:

$$E(\theta \mid y) \approx \frac{1}{M} \sum_{m=1}^M \theta^{(m)},$$

$$P(\theta > 0 \mid y) \approx \frac{1}{M} \sum_{m=1}^M I(\theta^{(m)} > 0).$$

Posterior quantiles give credible intervals.

For a future observation $\tilde{Y}$,

$$p(\tilde{y} \mid y) = \int p(\tilde{y} \mid \theta) p(\theta \mid y) d\theta.$$

Thus posterior simulation can turn difficult posterior integrals into averages over simulated draws. The remaining question is how to generate such draws when direct sampling is difficult.

Interactive 1: derive a likelihood

Suppose

$$Y_1, \dots, Y_n \mid \lambda \stackrel{iid}{\sim} \text{Poisson}(\lambda).$$

In pairs:

1. Write the pmf for one observation.
2. Multiply the terms to obtain $L(\lambda; y)$.
3. Remove factors not involving λ .
4. Write the log-likelihood.
5. With prior $\lambda \sim \text{Gamma}(a, b)$, identify the posterior kernel.

Do not start by remembering a conjugate formula. Start from the sampling model.

Interactive 1: solution

1. Start from the pmf for one observation.

For $Y_i \mid \lambda \sim \text{Poisson}(\lambda)$,

$$p(y_i \mid \lambda) = \frac{e^{-\lambda} \lambda^{y_i}}{y_i!}.$$

2. Use independence to form the likelihood.

$$L(\lambda; y) = \prod_{i=1}^n p(y_i \mid \lambda) = \prod_{i=1}^n \frac{e^{-\lambda} \lambda^{y_i}}{y_i!} = e^{-n\lambda} \lambda^{\sum_i y_i} \prod_{i=1}^n \frac{1}{y_i!}.$$

3. Remove factors not involving λ .

$$L(\lambda; y) \propto e^{-n\lambda} \lambda^{\sum_i y_i}.$$

4. Take logs.

Up to an additive constant,

$$\ell(\lambda) = -n\lambda + \left(\sum_i y_i \right) \log \lambda.$$

Interactive 1: solution – posterior

5. Add the prior and identify the posterior kernel.

For a Gamma prior in shape–rate parameterization,

$$p(\lambda) \propto \lambda^{a-1} e^{-b\lambda}.$$

Therefore

$$p(\lambda \mid y) \propto L(\lambda; y)p(\lambda)$$

and hence

$$p(\lambda \mid y) \propto \lambda^{a+\sum_i y_i-1} e^{-(b+n)\lambda}.$$

This is the kernel of

$$\lambda \mid y \sim \text{Gamma} \left(a + \sum_i y_i, b + n \right).$$

The workflow is

sampling model $\rightarrow$ likelihood $\rightarrow$ prior $\rightarrow$ posterior kernel.

Why MCMC?

When direct posterior sampling is difficult

In simple conjugate models we can derive the posterior and sample from it directly.

In realistic models:

- ▶ the normalizing constant may be unavailable;
- ▶ the posterior may be multivariate;
- ▶ parameters may be strongly dependent;
- ▶ posterior expectations may have no closed form.

We often want

$$E\{h(\theta) \mid y\} = \int h(\theta)p(\theta \mid y) d\theta.$$

If independent posterior draws are unavailable, we need another way to explore the posterior.

The goal: Markov chain Monte Carlo

Markov chain Monte Carlo (MCMC) constructs a sequence

$$\theta^{(1)}, \theta^{(2)}, \dots$$

that is generally **dependent**, but whose long-run behavior is governed by the target posterior

$$\pi(\theta) = p(\theta \mid y).$$

Then posterior expectations can be approximated by time averages:

$$E_{\pi}\{h(\theta)\} \approx \frac{1}{M} \sum_{m=1}^M h(\theta^{(m)}).$$

Before constructing such algorithms, we need the concepts that make this possible:

- ▶ Markov chains;
- ▶ stationary distributions;
- ▶ irreducibility and periodicity;
- ▶ convergence of distributions versus convergence of averages.

Markov chains

A stochastic process $X_0, X_1, \dots$ is a Markov chain if

$$P(X_{t+1} \in A \mid X_t, X_{t-1}, \dots, X_0) = P(X_{t+1} \in A \mid X_t).$$

The next state depends on the past only through the current state.

A transition kernel $P(x, A)$ specifies the probability of moving from state x into set A .

For a finite state space, this is represented by a transition matrix

$$P = (p_{ij}), \quad p_{ij} = P(X_{t+1} = j \mid X_t = i).$$

Stationary distribution

A distribution π is stationary if

$$\pi(A) = \int P(x, A) \pi(dx).$$

For a finite chain,

$$\pi^\top P = \pi^\top.$$

If

$$X_t \sim \pi,$$

then also

$$X_{t+1} \sim \pi.$$

In MCMC we design P so that the desired posterior π is stationary.

Detailed balance

A transition kernel satisfies **detailed balance** with respect to π if

$$\pi(x)P(x, dy) = \pi(y)P(y, dx).$$

For a finite state space this means

$$\pi_i p_{ij} = \pi_j p_{ji} \quad \text{for every pair } i, j.$$

Interpretation: when the chain is in equilibrium, the probability flow from i to j equals the probability flow from j to i .

A chain satisfying detailed balance is also called **reversible with respect to** π .

Why does detailed balance imply stationarity?

For a finite chain, detailed balance gives

$$\pi_i p_{ij} = \pi_j p_{ji}.$$

The probability of being in state j after one transition is

$$\sum_i \pi_i p_{ij}.$$

Using detailed balance,

$$\sum_i \pi_i p_{ij} = \sum_i \pi_j p_{ji} = \pi_j \sum_i p_{ji} = \pi_j.$$

Thus

$$\boxed{\pi^\top P = \pi^\top.}$$

So detailed balance implies stationarity.

Why is detailed balance stronger than stationarity?

Stationarity requires only that the **total flow into each state** has the correct value:

$$\sum_i \pi_i p_{ij} = \pi_j.$$

Detailed balance requires the stronger pairwise conditions

$$\pi_i p_{ij} = \pi_j p_{ji} \quad \text{for every } i, j.$$

Therefore

$$\boxed{\text{detailed balance} \implies \text{stationarity},}$$

but generally

$$\boxed{\text{stationarity} \not\implies \text{detailed balance}.}$$

Stationarity permits a net circulation of probability around the state space; detailed balance rules out such pairwise net flow.

Example: stationary but not reversible

Consider

$$P = \begin{pmatrix} 0.1 & 0.9 & 0 \\ 0 & 0.1 & 0.9 \\ 0.9 & 0 & 0.1 \end{pmatrix}.$$

The chain tends to move

$$1 \rightarrow 2 \rightarrow 3 \rightarrow 1.$$

The uniform distribution

$$\pi = \left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3} \right)$$

is stationary because $\pi^\top P = \pi^\top$. But

$$\pi_1 p_{12} = \frac{1}{3}(0.9) = 0.3, \quad \pi_2 p_{21} = \frac{1}{3}(0) = 0.$$

Hence detailed balance fails: the chain is stationary but not reversible.

Why use detailed balance in MCMC?

Detailed balance is **not necessary** for a valid MCMC algorithm.

It is useful because it provides a convenient way to prove that the desired target distribution is stationary.

Instead of proving directly that

$$\int \pi(dx) P(x, dy) = \pi(dy),$$

we establish the stronger relation

$$\pi(dx) P(x, dy) = \pi(dy) P(y, dx).$$

Metropolis–Hastings uses exactly this strategy: its acceptance probability is constructed so that detailed balance holds.

Thus detailed balance is best viewed as a convenient **construction and proof principle**.

Stationarity is not enough

Consider the extreme transition matrix

$$P = I.$$

Then

$$\pi^\top P = \pi^\top$$

for **every** distribution π .

But if $X_0 = x$, then

$$X_t = x \quad \text{for every } t.$$

The chain never explores the target distribution.

Thus stationarity tells us what happens **if the chain is already at equilibrium**. For MCMC we also need conditions ensuring that the chain can reach and explore equilibrium from its starting point.

Irreducibility: can the chain explore?

Informally, a chain is **irreducible** if every relevant part of the state space can eventually be reached from every other relevant part.

For a finite chain, for every pair i, j , there must exist some $t \geq 1$ such that

$$P^t(i, j) > 0.$$

For example,

$$P = \begin{pmatrix} 0.8 & 0.2 & 0 \\ 0.2 & 0.8 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

is not irreducible. States 1 and 2 cannot communicate with state 3.

For MCMC, lack of irreducibility means that initialization may determine which part of the target the chain can ever visit.

Aperiodicity: avoiding deterministic cycles

Even an irreducible chain can fail to converge from a fixed starting point. Consider

$$P = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Starting from state 1,

$$1 \rightarrow 2 \rightarrow 1 \rightarrow 2 \rightarrow \dots$$

The chain can return to state 1 only after 2, 4, 6, ... steps, so its period is 2.

It has stationary distribution $\pi = (1/2, 1/2)$, but $P(X_t = 1)$ alternates between 1 and 0 instead of converging to $1/2$.

Aperiodicity excludes this kind of forced cyclic behavior **of the distribution**.

Aperiodicity: avoiding deterministic cycles

But notice something important: although $P(X_t = 1)$ does not converge, the time average does:

$$\frac{1}{M} \sum_{t=1}^M I(X_t = 1) \longrightarrow \frac{1}{2}.$$

For this finite irreducible periodic chain, visits alternate between the two states, so approximately half of the observations are in state 1.

Thus periodicity can prevent $P^t(x, \cdot)$ from converging while long-run averages still converge.

What do we mean by ergodicity?

The term **ergodic** has somewhat different technical formulations in different Markov-chain settings.

For this course, the important idea is:

An ergodic chain eventually forgets its starting point and explores its stationary distribution sufficiently well for long-run averages to represent expectations under that distribution.

There are two related properties that we should keep separate.

For a finite **irreducible and aperiodic** chain with stationary distribution π , $P^t(x, \cdot) \longrightarrow \pi(\cdot)$.

Thus the distribution of a late state X_t approaches π .

For a finite **irreducible** chain, even if it is periodic, time averages satisfy an ergodic law:

$$\frac{1}{M} \sum_{t=1}^M h(X_t) \longrightarrow E_{\pi}\{h(X)\}.$$

So aperiodicity is needed for convergence of the distribution from a fixed start, but not for this finite-state law of large numbers.

Two different convergence statements

It is essential to distinguish these statements.

1. Convergence of the distribution:

$$P^t(x, \cdot) \longrightarrow \pi(\cdot).$$

For a finite irreducible chain, **aperiodicity is additionally required** for this convergence from a fixed starting state.

2. Convergence of averages:

$$\frac{1}{M} \sum_{t=1}^M h(X_t) \longrightarrow E_{\pi}\{h(X)\}.$$

For a finite irreducible chain, this time-average result holds even when the chain is periodic.

The second statement is the one that ultimately justifies Monte Carlo estimation from a Markov-chain trajectory.

Markov chain Monte Carlo estimator

Suppose the chain has stationary distribution π . We estimate

$$\mu_h = E_{\pi}\{h(X)\}$$

by

$$\hat{\mu}_M = \frac{1}{M} \sum_{t=1}^M h(X_t).$$

Under suitable ergodicity conditions,

$$\hat{\mu}_M \rightarrow \mu_h.$$

This resembles ordinary Monte Carlo, but the draws are dependent.

Dependence and Monte Carlo variance

For a stationary chain, define

$$\sigma_h^2 = \text{Var}_\pi\{h(X)\}$$

and let

$$\rho_k = \text{Corr}\{h(X_t), h(X_{t+k})\}.$$

Under a Markov chain CLT,

$$\text{Var}(\hat{\mu}_M) \approx \frac{\sigma_h^2}{M} \left(1 + 2 \sum_{k=1}^{\infty} \rho_k \right).$$

Define the integrated autocorrelation time

$$\tau_{\text{int}} = 1 + 2 \sum_{k=1}^{\infty} \rho_k.$$

Then the effective sample size is approximately

$$M_{\text{eff}} = \frac{M}{\tau_{\text{int}}}.$$

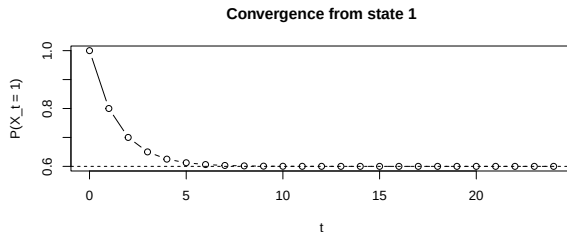
Ten thousand MCMC draws are not necessarily equivalent to ten thousand iid draws.

Transition matrices: one step versus many steps

If the initial distribution is the row vector $v_0^\top$, then

$$v_t^\top = v_0^\top P^t.$$

Thus convergence of a finite Markov chain can be studied through powers of P .



The horizontal line is the stationary probability $\pi_1 = 0.6$.

Conditions: what problem does each one solve?

For finite-state MCMC, the roles can be summarized as follows:

- ▶ **Stationarity:** π is preserved by the transition rule.
- ▶ **Irreducibility:** the chain can reach all relevant parts of the state space.
- ▶ **Aperiodicity:** together with irreducibility, ensures convergence of $P^t(x, \cdot)$ to π rather than persistent cycling.
- ▶ **Time-average convergence:** for a finite irreducible chain, averages recover expectations under π even if the chain is periodic.

A useful mental picture is

stationarity + irreducibility $\Rightarrow$ correct long-run averages

while aperiodicity additionally gives

convergence of the distribution from a fixed start.

For general-state-space MCMC the precise theory is more subtle, but these finite-state ideas explain the purpose of the conditions encountered later in Metropolis–Hastings and Gibbs sampling.

Ergodic theorem versus iid law of large numbers

For iid Monte Carlo,

$$X_1, \dots, X_M \stackrel{iid}{\sim} \pi$$

and the ordinary law of large numbers justifies averages.

For MCMC, the observations are dependent. Under suitable ergodicity conditions,

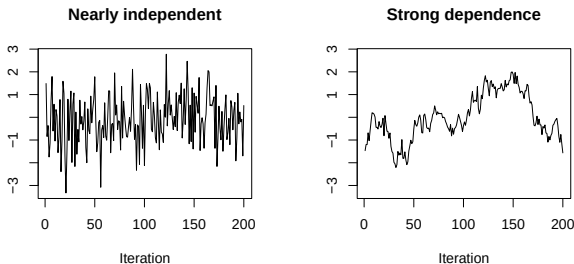
$$\frac{1}{M} \sum_{t=1}^M h(X_t) \longrightarrow E_{\pi}\{h(X)\}.$$

The key distinction is:

MCMC does not need independent draws, but it needs the right long-run behavior.

Autocorrelation and effective sample size: picture

The following two displayed series each contain 200 values with approximately standard-normal marginal behavior, but one is strongly autocorrelated.



The second sequence contains much less information about its mean despite having the same number of stored values.

A tiny Markov chain in R

```
P <- matrix(
  c(0.8, 0.2,
    0.3, 0.7),
  nrow = 2,
  byrow = TRUE
)
simulate_chain <- function(P, x0, n) {
  x <- integer(n)
  x[1] <- x0
  for (t in 2:n) {
    x[t] <- sample(
      seq_len(nrow(P)),
      size = 1,
      prob = P[x[t - 1], ]
    )
  }
  x
}
```

A tiny Markov chain in R

```
x <- simulate_chain(P, x0 = 1, n = 10000)
prop.table(table(x))
```

```
## x
##      1      2
## 0.5896 0.4104
```

The stationary probabilities solve

$$\pi^\top P = \pi^\top.$$

Interactive 2: stationary distribution

Consider

$$P = \begin{pmatrix} 0.8 & 0.2 \\ 0.3 & 0.7 \end{pmatrix}.$$

1. Let $\pi = (p, 1 - p)$. Solve $\pi^\top P = \pi^\top$.
2. Predict the long-run fraction of time spent in each state.
3. Run the R simulation from the previous slide.
4. Change the initial state. Does the long-run result change?
5. What would go wrong if

$$P = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}?$$

Interactive 2: solution

Let

$$\pi = (p, 1 - p).$$

Stationarity requires

$$\pi^\top P = \pi^\top.$$

From the first component,

$$p = 0.8p + 0.3(1 - p) = 0.5p + 0.3,$$

so

$$0.5p = 0.3, \quad p = 0.6.$$

Thus

$$\boxed{\pi = (0.6, 0.4)}.$$

The predicted long-run fractions of time in states 1 and 2 are therefore 0.6 and 0.4.

Interactive 2: solution – both starting states

```
set.seed(20260812)

x1 <- simulate_chain(P, x0 = 1, n = 10000)
x2 <- simulate_chain(P, x0 = 2, n = 10000)

rbind(
  start_1 = prop.table(table(factor(x1, levels = 1:2))),
  start_2 = prop.table(table(factor(x2, levels = 1:2)))
)
```

```
##           1      2
## start_1 0.5862 0.4138
## start_2 0.6024 0.3976
```

Both empirical distributions should be close to $(0.6, 0.4)$. The starting state influences the beginning of the trajectory, but not the long-run proportions for this irreducible chain.

Interactive 2: solution – both starting states

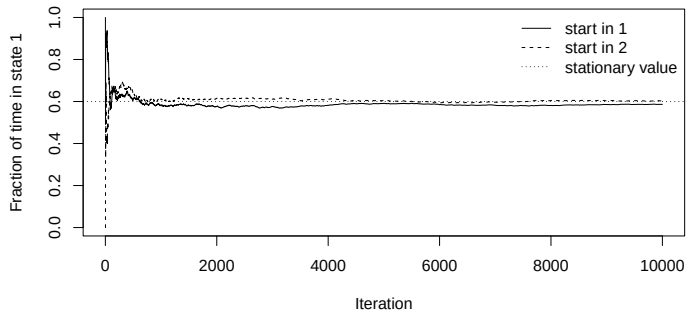
The convergence of the time averages can be seen directly:

```
r1 <- cumsum(x1 == 1) / seq_along(x1)
r2 <- cumsum(x2 == 1) / seq_along(x2)

plot(r1, type = "l", ylim = c(0, 1),
     xlab = "Iteration",
     ylab = "Fraction of time in state 1")
lines(r2, lty = 2)
abline(h = 0.6, lty = 3)

legend(
  "topright",
  c("start in 1", "start in 2", "stationary value"),
  lty = c(1, 2, 3), bty = "n"
)
```

Interactive 2: solution – both starting states



Both running proportions approach the stationary probability $\pi_1 = 0.6$.

Interactive 2: solution – identity chain

Now consider

$$P_I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

Every distribution $\pi = (p, 1 - p)$ is stationary because

$$\pi^\top P_I = \pi^\top.$$

But the chain never changes state.

```
P_identity <- diag(2)
z1 <- simulate_chain(P_identity, x0 = 1, n = 20)
z2 <- simulate_chain(P_identity, x0 = 2, n = 20)
z1
```

```
## [1] 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
```

```
z2
```

```
## [1] 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2
```

Interactive 2: solution – identity chain

Starting in state 1 gives $1, 1, 1, \dots$, while starting in state 2 gives $2, 2, 2, \dots$

The problem is that the identity chain is **not irreducible**: the two states do not communicate. Its long-run empirical distribution therefore depends completely on the initial state.

This separates two ideas:

- ▶ **stationarity**: a distribution is unchanged by a transition;
- ▶ **ergodic exploration**: the chain moves through the relevant state space so that time averages recover target expectations.

Hence

having the desired stationary distribution is not enough.

From Markov chains to MCMC

The MCMC design problem

We know the target density only up to proportionality:

$$\pi(\theta) \propto \tilde{\pi}(\theta).$$

Computationally, we will normally work with

$$\log \tilde{\pi}(\theta) = \log p(y \mid \theta) + \log p(\theta) + C,$$

rather than multiplying many small densities. This links directly back to the floating-point issues from Lecture 1: products of likelihood contributions can underflow, whereas sums of log densities are numerically much more stable.

We need to construct a transition rule that:

1. can be simulated;
2. leaves π invariant;
3. can move throughout the relevant state space;
4. mixes sufficiently rapidly for Monte Carlo estimation.

Next lecture: **Metropolis–Hastings** provides a general solution.

Summary

- ▶ Bayes combines likelihood and prior:

$$p(\theta \mid y) \propto p(y \mid \theta)p(\theta).$$

- ▶ Writing the likelihood starts from the sampling model and independence assumptions.
- ▶ MCMC replaces difficult direct sampling by a Markov chain with the target posterior as stationary distribution.
- ▶ Stationarity alone is insufficient; exploration and convergence matter.
- ▶ Autocorrelation reduces effective sample size.
- ▶ The central computational question is:

How can we construct a chain whose stationary distribution is the target?